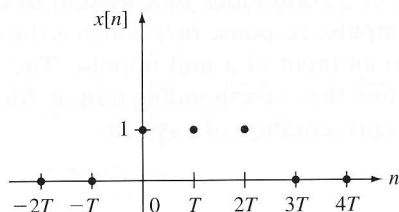
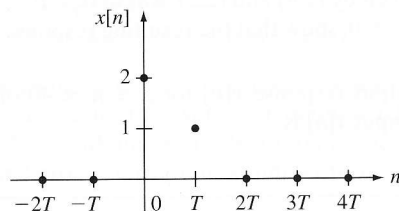
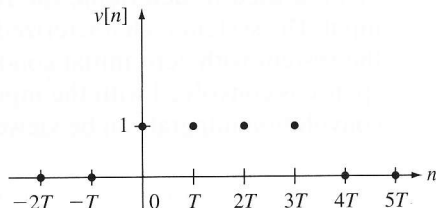


(d) $y[n+2] + 1/2y[n] = 2x[n+2] - x[n]$

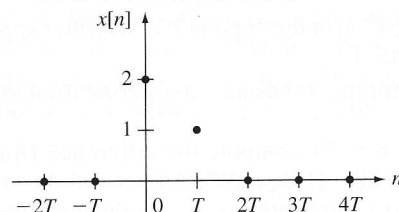
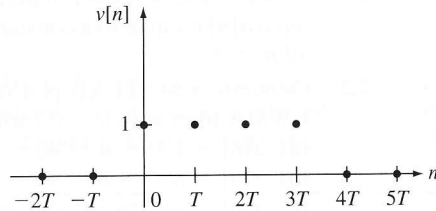
2.7. For the discrete-time signals $x[n]$ and $v[n]$ shown in Figure P2.7, do the following:



(a)



(b)



(c)

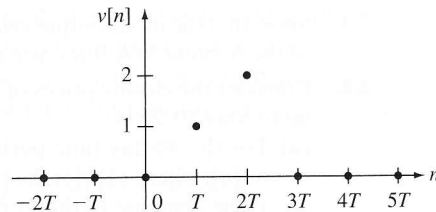


FIGURE P2.7

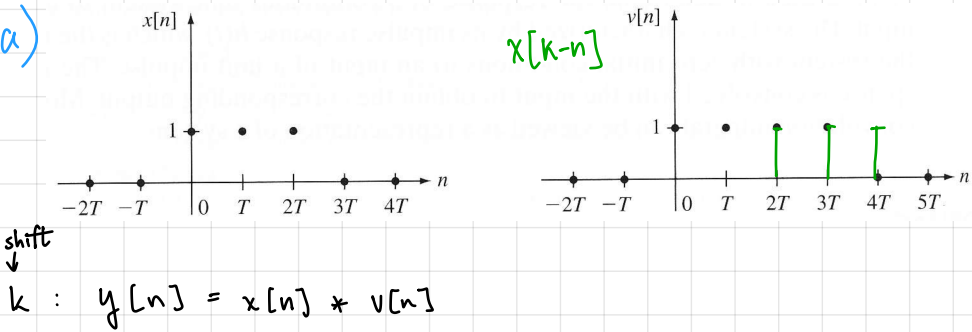
(a) Compute the convolution $x[n] * v[n]$ for all $n \geq 0$. Sketch the results.

(b) Repeat part (a), using the M-file `conv`.

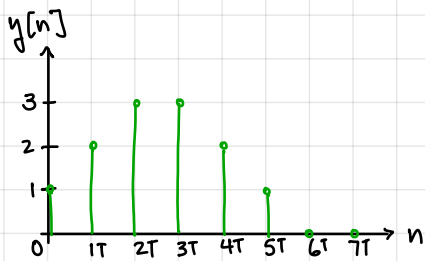
2.8 For the discrete-time signals $x[n]$ and $v[n]$ given in each of the following parts, compute

2.7)

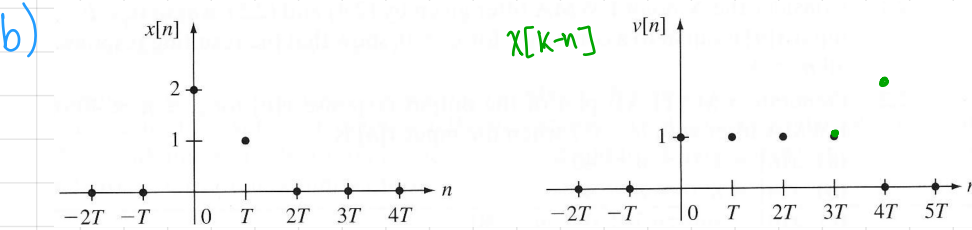
a)



- 0 $y[0] = 1$
- 1 $y[1] = 2$
- 2 $y[2] = 3$
- 3 $y[3] = 3$
- 4 $y[4] = 2$
- 5 $y[5] = 1$
- 6 $y[6] = 0$

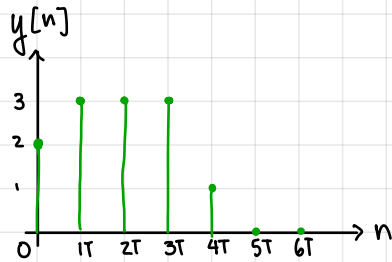


b)

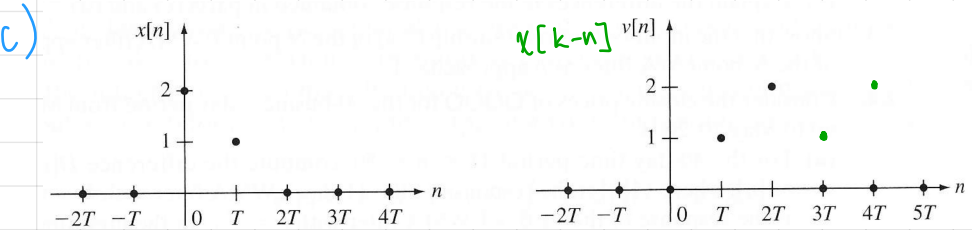


k = shift $y[n] = x[n] * v[n]$

- k = 0 $y[0] = 2 \cdot 1 = 2$
- k = 1 $y[1] = 1 \cdot 1 + 2 \cdot 1 = 3$
- k = 2 $y[2] = 1 \cdot 1 + 2 \cdot 1 = 3$
- k = 3 $y[3] = 1 \cdot 1 + 2 \cdot 1 = 3$
- k = 4 $y[4] = 1 \cdot 1 = 1$
- k = 5 $y[5] = 0$



c)



k = shift $y[n] = x[n] * v[n]$

- k = 0 $y[0] = 0$
- k = 1 $y[1] = 1 \cdot 2 = 2$
- k = 2 $y[2] = 1 \cdot 1 + 2 \cdot 2 = 5$
- k = 3 $y[3] = 1 \cdot 2 = 2$
- k = 4 $y[4] = 0$

